

Task 1: Formulating your world as an MDP

1. MDP Design

Scenario: Deciding whether to study or relax on campus in the evening.

State Space (S):

s1: Hostel

s2: Library

s3: OAT

Action Space (A):

a1: Study

a2: Relax

Transition Probabilities (P):

From Hostel (s1):

Study (a1): $P(s2 | s1, a1) = 0.8$, $P(s1 | s1, a1) = 0.2$

Relax (a2): $P(s3 | s1, a2) = 0.9$, $P(s1 | s1, a2) = 0.1$

From Library (s2):

Study (a1): $P(s2 | s2, a1) = 1.0$

Relax (a2): $P(s3 | s2, a2) = 0.7$, $P(s1 | s2, a2) = 0.3$

From OAT (s3):

Study (a1): $P(s2 | s3, a1) = 0.6$, $P(s1 | s3, a1) = 0.4$

Relax (a2): $P(s3 | s3, a2) = 1.0$

Rewards (R):

Hostel (s1): $R(s1, a1) = 2$, $R(s1, a2) = 5$

Library (s2): $R(s2, a1) = 10$, $R(s2, a2) = -2$

OAT (s3): $R(s3, a1) = -2$, $R(s3, a2) = 8$

2. Manual Calculation: Value Iteration

Discount Factor: $\gamma = 0.9$

Update Rule: $V_{\text{next}}(s) = \max [R(s, a) + \gamma * (\text{sum of } P * V_{\text{current}})]$

Initialization ($k = 0$)

$$V_0(s_1) = 0$$

$$V_0(s_2) = 0$$

$$V_0(s_3) = 0$$

Iteration 1 ($k = 1$)

Since the previous values (V_0) are all 0, we only look at the immediate rewards:

$$\text{Hostel } (s_1): V_1(s_1) = \max(2, 5) = 5$$

$$\text{Library } (s_2): V_1(s_2) = \max(10, -2) = 10$$

$$\text{OAT } (s_3): V_1(s_3) = \max(-2, 8) = 8$$

Iteration 2 ($k = 2$)

For Hostel (s_1):

$$\text{Study: } 2 + 0.9 * (0.2 * 5 + 0.8 * 10) = 2 + 0.9 * (1 + 8) = 10.1$$

$$\text{Relax: } 5 + 0.9 * (0.1 * 5 + 0.9 * 8) = 5 + 0.9 * (0.5 + 7.2) = 11.93$$

$$V_2(s_1) = \max(10.1, 11.93) = 11.93$$

For Library (s_2):

$$\text{Study: } 10 + 0.9 * (1.0 * 10) = 10 + 9 = 19.0$$

$$\text{Relax: } -2 + 0.9 * (0.3 * 5 + 0.7 * 8) = -2 + 0.9 * (1.5 + 5.6) = 4.39$$

$$V_2(s_2) = \max(19.0, 4.39) = 19.0$$

For OAT (s_3):

$$\text{Study: } -2 + 0.9 * (0.4 * 5 + 0.6 * 10) = -2 + 0.9 * (2 + 6) = 5.2$$

$$\text{Relax: } 8 + 0.9 * (1.0 * 8) = 8 + 7.2 = 15.2$$

$$V_2(s_3) = \max(5.2, 15.2) = 15.2$$

Final Values After 2 Steps

$$V_2(\text{Hostel}) = 11.93$$

$$V_2(\text{Library}) = 19$$

$$V_2(\text{OAT}) = 15.2$$